RunSoC 2.0
Research Methods Protocol

Artifact for the Research Methods

1. Purpose and Scope

Purpose. This document records the review-protocol context for the RunSoC 2.0 research method. It is intended as a compact, auditable methods artifact as an overview of the rapid literature review design, search strategy, and data extraction plan.

Scope. The review concerns semiconductor design partitions in heterogeneous multiprocessor systems-on-chip (MPSoCs) and the allocation of automotive runnables/tasks to such partitions in mixed-criticality high-performance computers (HPCs).

2. Research Questions

The final research-question set is:

- **RQ1:** Which types of semiconductor design partitions are used in heterogeneous MPSoCs and what are their affinities/properties?
- **RQ2:** How can automotive runnables/tasks be allocated within automotive, heterogeneous HPC MPSoCs based on affinity?

Positioning note. The research questions are formulated independently of a specific automotive function domain

3. PICOC Criteria

The rapid review is structured using PICOC criteria derived from systematic-literature-review guidance for software engineering.

PICOC Element	Definition for RunSoC 2.0	Role in Review Design
Population	Semiconductor design partitions in heterogeneous MPSoCs.	Defines the hardware entities to be identified and characterized, such as CPU clusters, real-time cores, accelerators, GPUs, NPUs, DSPs, memory/security islands, and other relevant SoC partitions.
Intervention	Allocation of runnables/tasks to semiconductor design partitions based on affinity.	Defines the design activity under investigation: mapping software functions to hardware partitions according to requirements and partition capabilities.
Comparison	Homogeneous multi-core microcontroller.	Provides the baseline architecture against which heterogeneous MPSoCs are conceptually contrasted.

PICOC Element	Definition for RunSoC 2.0	Role in Review Design
Outcomes	Overview of automotive semiconductor design partitions and properties; framework for allocating runnables/tasks to heterogeneous SoCs.	Specifies the expected review products: a taxonomy/properties overview and allocation-support framework.
Context	Automotive mixed-criticality HPCs.	Restricts interpretation to automotive high-performance computing platforms that host functions with differing safety, timing, isolation, and performance requirements.

4. Review Protocol Context

Protocol Item	Specification
Review type	Rapid literature review complemented by semi-structured expert talks/interviews.
Review objective	Identify the relevant semiconductor design partitions in automotive heterogeneous MPSoCs and extract properties that influence runnable/task allocation.
Search strategy	Hybrid search strategy combining database search and iterative refinement. Candidate databases: IEEE Xplore and/or Scopus.
Selection process	Search results should be screened using explicit inclusion and exclusion criteria to reduce subjective selection risk.
Data extraction	Extract hardware-partition types, properties, software/task requirements, allocation criteria, automotive context, and evidence type.
Data synthesis	Synthesize findings thematically; candidate outputs include taxonomy tables, property matrices, allocation-criterion mappings, and summary charts.
Expert input	Expert talks at Daimler Truck AG are used to validate domain relevance, clarify architecture terminology, and refine allocation criteria.

5. Final Search String

Candidate database query, formulated for Scopus-style TITLE-ABS-KEY syntax:

```
TITLE-ABS-KEY( (automotive) AND ("System-on-Chip" OR SoC OR "heterogeneous multicore" OR
"heterogeneous processor" OR "heterogeneous platform" OR "multi-core architecture" OR
MPSoC) AND ( (”core affinity” OR affinity) OR (allocation OR mapping OR scheduling OR
"task allocation" OR "software allocation") ) AND NOT (battery OR charging OR powertrain-
only))) → 107 results
```

6. Inclusion/Exclusion Criteria

Inclusion
IC1: the paper presents or discusses a taxonomy or categorization of automotive, heterogeneous SoCs or their partitions
IC2: the paper presents or discusses properties and affinities of automotive, heterogeneous SoCs or their partitions
IC3: the paper presents or discusses a framework or algorithm for task allocation to automotive, heterogeneous SoC partitions
Exclusion
EC1: the paper does not meet any of the inclusion criteria
EC2: the paper is duplicated
EC3: the paper is not peer-reviewed
EC4: the paper is either a book or a chapter from a book
EC5: the paper is not written in English or German
EC6: the paper does not have available full text
EC7: Secondary studies/ Conference proceedings

7. Data Extraction Fields

Field Group	Example Candidates

DEC1: Which type of heterogeneous MPSoCs and partitions does the study elaborate on?	Xilinx Zynq UltraScale+ MPSoC ZCU102
	• Partitions such as:
	  ○ GPU
	  ○ DSP
	  ○ FPGA
	  ○ ASIC / accelerator
	  ○ Memory subsystem
	  ○ NoC

DEC2: Which heterogeneous MPSoC partitions are discussed and what are their properties/affinities?	• Performance • Power consumption • Latency • Determinism • Safety criticality • Parallelism • Memory bandwidth • Thermal constraints

Field Group	Example Candidates
DEC3: Which affinities between runnables/tasks and hardware partitions?	See DEC2
DEC4: Which are the key challenges in allocating runnables to heterogeneous MPSoCs?	• Timing behavior • Scalability • Complexity • Determinism • Isolation • Safety support • Performance • Energy • Memory locality • Toolchain support.
DEC5: Which methodologies or frameworks are used to allocate runnables to heterogeneous SoCs?	• Heuristic-based • Optimization-based (ILP, MILP, etc.) • Rule-based • Model-based (e.g., AUTOSAR, UML/MARTE) • Machine learning–based • Custom framework • Conceptual (no implementation)
DEC6: Which validation methods or case studies are presented to demonstrate the effectiveness of the proposed methodology or framework?	• Simulation • Analytical evaluation • Case study • Industrial use case • Benchmark
DEC7: Are any industry standards or best practices followed in the allocation process?	• AUTOSAR (Classic / Adaptive) • ISO 26262 • ASPICE • POSIX
DEC8: What are the benefits / targets of the proposed method?	See DEC4